

# Digital Circuits and VHDL – Overview and Examples

Florian Hagemann

*Abstract—Here be an abstract...*

## CONTENTS

|            |                                          |   |
|------------|------------------------------------------|---|
| <b>I</b>   | <b>Introduction</b>                      | 1 |
| <b>II</b>  | <b>Digital Circuits</b>                  | 1 |
| II-A       | Electronical Switches: Transistors . . . | 1 |
| II-B       | Making Computers Think . . . . .         | 1 |
| II-C       | Examples . . . . .                       | 1 |
| <b>III</b> | <b>Introduction to VHDL</b>              | 1 |
| III-A      | Hardware Description Languages . . . .   | 1 |
| III-B      | Basic Syntax . . . . .                   | 1 |
| III-C      | Blackbox Diagrams and Entities . . . .   | 1 |
| <b>IV</b>  | <b>Combinatorial Circuits</b>            | 1 |
| IV-A       | . . . . .                                | 1 |
| <b>V</b>   | <b>Sequential Circuits</b>               | 1 |
| <b>VI</b>  | <b>Finite State Machines (FSM)</b>       | 1 |
| VI-A       | Moore Machines . . . . .                 | 1 |
| VI-B       | Mealy Machines . . . . .                 | 1 |
|            | <b>References</b>                        | 1 |

*A. Electronical Switches: Transistors*

*B. Making Computers Think*

*C. Examples*

## III. INTRODUCTION TO VHDL

*A. Hardware Description Languages*

*B. Basic Syntax*

*C. Blackbox Diagrams and Entities*

## IV. COMBINATORICAL CIRCUITS

*A.*

## V. SEQUENTIAL CIRCUITS

## VI. FINITE STATE MACHINES (FSM)

*A. Moore Machines*

*B. Mealy Machines*

## REFERENCES

- [1] Wiktionary contributors, circuit. Wiktionary, the free dictionary. [Online]. Available: <https://en.wiktionary.org/wiki/circuit>
- [2] —, digital. Wiktionary, the free dictionary. [Online]. Available: <https://en.wiktionary.org/wiki/digital>
- [3] B. Mealy and F. Tappero, *Free Range VHDL*. Free Range Factory, 2025.
- [4] D. M. Harris and S. L. Harris, *Digital Design and Computer Architecture: RISC-V Edition*, 2nd ed. Elsevier, 2022.
- [5] M. M. Mano and M. D. ciletti, *Digital Design*, 5th ed. Pearson, 2013.

## I. INTRODUCTION

## II. DIGITAL CIRCUITS

**W**HAT is a *digital circuit*? To understand this, we need to examine each word separately. A *circuit* is a closed loop that allows electricity to flow. Typically it consists of a power source and some device (e.g. a motor, LED, etc.) that uses that electricity. A circuit needs to be closed in order for the electricity to flow through it [1]. *Digital* comes from the Latin "*digitalis*" meaning finger or toe. Digital can mean "something to do with digits (fingers or toes)" but can also mean the "property of representing values as discrete, often binary, numbers rather than a continuous spectrum" [2].

**A digital circuit**

Thanks to ...